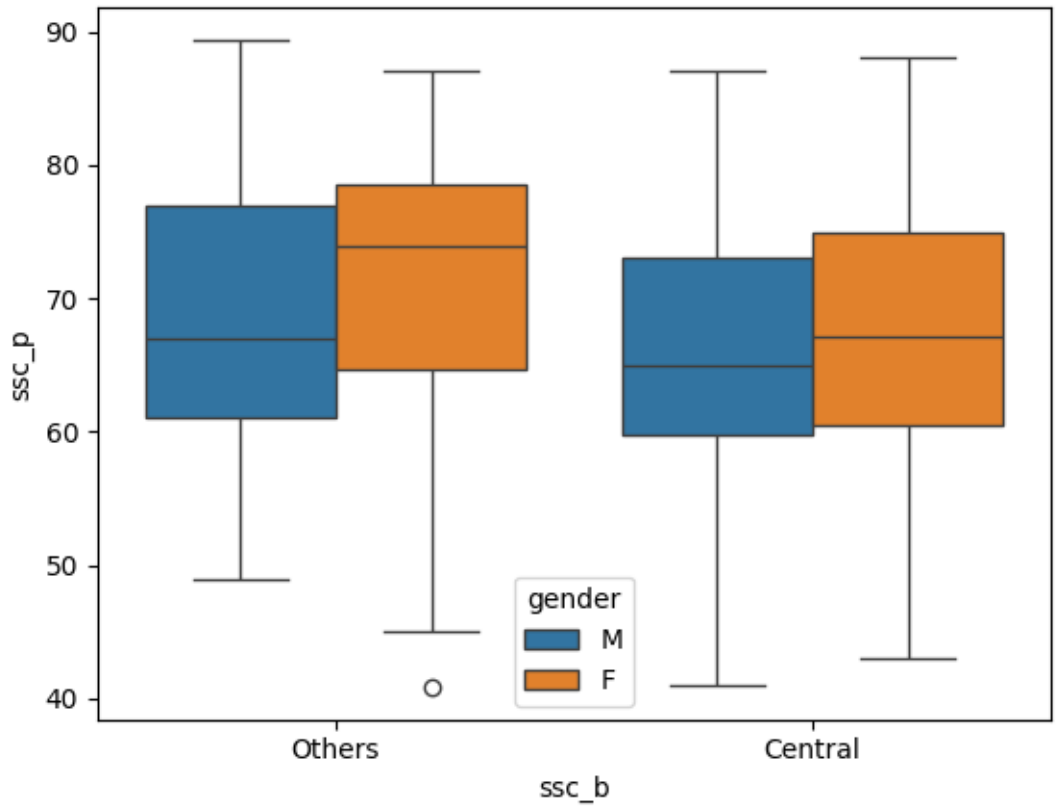
Box plot

A **boxplot** is a statistical graph that shows the **distribution of a dataset** based on five summary statistics:

1. **Minimum** (lowest non-outlier value)
2. **Q1 (First Quartile, 25th percentile)**
3. **Median (Q2, 50th percentile)**
4. **Q3 (Third Quartile, 75th percentile)**
5. **Maximum** (highest non-outlier value)



Summary of box plot:

Axes:

- Y-axis (ssc_p) → Student's secondary school percentage (scores).

- X-axis (ssc_b) → Type of secondary education board:
  - Others
  - Central
- Colors → Gender:
  - Blue = Male (M)
  - Orange = Female (F)

Interpretation of the boxplots:

1. **Others Board**

- Male (M, blue):
  - Median ≈ 67%
  - Middle 50% (IQR) between ~61% and ~77%
  - Range: ~49% to ~89%
  - No outlier
- Female (F, orange):
  - Median ≈ 74% (higher than males)
  - Middle 50% between ~65% and ~78%
  - Range: ~45% to ~87%
  - One outlier at ~41%

Observation: In the "Others" board, **females performed better** than males on average (**higher median and spread).**

2. **Central Board**

- Male (M, blue):
  - Median ≈ 65%

    - IQR between ~60% and ~73%
    - Range: ~41% to ~87%
- Female (F, orange):
    - Median ≈ 67% (slightly higher than males, but difference is small)
    - IQR between ~61% and ~76%
    - Range: ~42% to ~88%

Observation: In the "Central" board, male and female scores are **quite similar**, with only a **small difference in medians.**

Overall Insights

1. Female students (orange) generally scored higher than male students, especially in the "Others" board.
2. Spread (IQR) is larger in the "Others" board for males
3. Outlier present: One female student in "Others" board scored unusually low (~41%).
4. Central vs Others: Median scores are slightly higher in "Others" (especially for females).